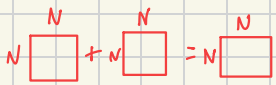
Meeting Notes June 9th

Roofline model

↙ another way of deciding how to perform

What is the least amount of data that can be moved?



$3N^2$

What is the best way to write a kernel?

↳ how do we write smallest piece of code that utilizes the most of the system →

↳ how can I use the machine the best to perform operations.

Scalar

$c += a \cdot b$ → one instruction, one output

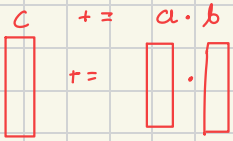
V= SIMD line

↓

single instruction → V # of values

In order to write SIMD, you have to rewrite code

↳ find the independent operations



$c += a_0 b_0 + a_1 b_1 + a_{v-1} b_{v-1}$

$C_0 \; r = a_0 b_0$

$C_i \; += a_i b_i$ ← this can be done through SIMD

Accumulation variable in our code

Pixels are doing same operation and can be done in any order } in innermost loop

independent

↓

not doing it currently

How can I restructure code so I can compute multiple pixels w/ the same instruction 3×3 blur NOT Separable

Write SIMD implementation → not code